

What is New in Multimeters

A multimeter for an electronics engineer is as essential as a stethoscope is for a physician. It is an extremely useful, indispensable diagnostic tool for electronics design professionals, students and hobbyists—but is there anything more to it? This article provides some insights into the latest trends in multimeters



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A multimeter is a very handy electronics measuring tool that helps in determining several electrical properties. Today, multimeters are available with advanced measuring capabilities for measuring capacitance in farads (F), inductance in henries (H), conductance in siemens (S), frequency in hertz (Hz), percentage duty cycle and also decibels (dB). With an appropriate test probe, such as a thermocouple, the temperature can also be determined in degree centigrade (°C) or Fahrenheit (°F).

The simplest electrical measurement testing tool, multimeter, has been evolving as a high-end electrical and electronics testing equipment to serve complex testing needs. Moreover, the latest multimeters incorporate a higher level of integration, connectivity and advanced features.

The trends

The multimeters are either hand-

held or bench-top type. Handheld multimeters are primarily used for insulation and other measurements in field. Bench-top multimeters are used for higher-accuracy applications in research laboratories and electronics design houses.

The benefits of digital multimeters (DMMs) based on the increasingly popular modular platform, especially PCI eXtensions for instrumentation (PXI), are also discussed in this article.

Wireless-enabled DMMs

Engineers and technicians often come across situations where they have to take readings from an operating machine, or from points high above the ground, or measure multiple points at the same time. Improving safety and productivity at work are very important. It appears that multimeters are evolving to meet these demands from the users.

Wireless technology is used to solve these issues, without compromising the safety of the user. It allows to

link distantly located multiple modules to a PC, tablet or smartphone. This allows remote logging of measurements in real time, and subsequently makes analysis and troubleshooting easier.

There are also Bluetooth-enabled multimeters available in the market. Wireless adaptors allowing log data from multiple DMMs in your computer/smartphone are also available now. Addition of these features provides an edge to the new multimeters over the traditional multimeters.

Multimeter with a remote display is another interesting product developed in this category. It allows detachment of the display from the body of the instrument and take readings from a remote location.

Software-defined instruments

Having highly scalable dense software architecture is very crucial for all the instruments. Hence in-line processing technology, such as field-programmable gate arrays (FPGAs) and real-time processors, are being adopted for multimeters. With the help of such technology, users can write an FPGA program to reconfigure the input/output (I/O) behaviour in order to define and design a particular multimeter as per requirement. Therefore, unlike the traditional multimeters, users can have a software-defined instrument rather than a vendor-defined one. The development time of a test system has been a great challenge and should be reduced considerably.

With regard to the wireless technology in the multimeter space, there are a growing number of software and mobile applications—some developed by vendors and the others by independent developers—to help analyse process